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Part I

Introduction

1 Background Knowledge

1.1 Markov Decision Process

Definition 1.1 (State, Action, Trajectory). A sequence of actions and states with T interactions with environment is defined as a trajectory $\tau = (s_0, a_0, s_1, a_1, s_2, a_2, \dots, a_{T-1}, s_{T-1})$

Definition 1.2 (Reward and Return). At time step t , the agent observes state s_t , takes action a_t , and transitions to the next state s_{t+1} . The single-step reward is determined by the transition and is defined as

$$r_t = R(s_t, a_t, s_{t+1}).$$

The (discounted) **reward** on the trajectory τ is defined as

$$R(\tau) = \sum_{t=0}^{T-1} \gamma^t r_t,$$

where $\gamma \in (0, 1]$ is the discount factor. The discounted **return** starting from time step t is defined as

$$G_t(\tau) = \sum_{t'=t}^{T-1} \gamma^{t'-t} r_{t'}.$$

Equivalently, G_t satisfies the recursive relation

$$G_t = r_t + \gamma G_{t+1}.$$

Definition 1.3 (State-Value Function and Value Function). The value function for the policy π is defined as

$$\begin{aligned} V^\pi(s_t) &= \mathbb{E}_\pi[G_t | s_t] = \mathbb{E}_\pi[r_t + \gamma G_{t+1} | s_t] = \mathbb{E}_{a_t \sim \pi(\cdot | s_t)} \mathbb{E}_{s_{t+1} \sim P(\cdot | s_t, a_t)} [r_t + \gamma \mathbb{E}_\pi[G_{t+1} | s_{t+1}]] \\ &= \mathbb{E}_{a_t \sim \pi(\cdot | s_t)} \mathbb{E}_{s_{t+1} \sim P(\cdot | s_t, a_t)} [r_t + \gamma V^\pi(s_{t+1})]. \end{aligned}$$

The state-value function for the policy π is defined as

$$Q^\pi(s_t, a_t) = \mathbb{E}_\pi[G_t | s_t, a_t] = \mathbb{E}_{s_{t+1} \sim P(\cdot | s_t, a_t)} [r_t + \gamma V^\pi(s_{t+1})].$$

Therefore,

$$V^\pi(s_t) = \mathbb{E}_{a_t \sim \pi(\cdot | s_t)} [Q^\pi(s_t, a_t)]$$

Definition 1.4 (Advantage). The advantage function, by its definition, measures whether or not the action is better or worse than the policy's default behavior.

$$A^\pi(s_t, a_t) = Q^\pi(s_t, a_t) - V^\pi(s_t) = \mathbb{E}_{s_{t+1} \sim P(\cdot | s_t, a_t)} \left[\underbrace{r_t + \gamma V^\pi(s_{t+1}) - V^\pi(s_t)}_{\text{temporal difference}} \right] \quad (1)$$

1.2 Useful Facts

1. Probability of a trajectory

$$P(\tau | \theta) = \rho_0(s_0) \prod_{t=0}^{T-1} P(s_{t+1} | s_t, a_t) \pi_\theta(a_t | s_t) \quad (2)$$

2. log derivative trick

$$\nabla_\theta P(\tau | \theta) = P(\tau | \theta) \nabla_\theta \log P(\tau | \theta) \quad (3)$$

3. expected Grad-Log-Prob Lemma

$$\mathbb{E}_{x \sim p(\cdot; \theta)} [\nabla_\theta \log p(x; \theta)] = 0 \quad (4)$$

1.3 Glossaries

Definition 1.5 (Estimation of the KL divergence). There are different estimators of the KL divergence, which can be found here. Recall that $\text{KL}(q||p) = \mathbb{E}_{x \sim q} \left[\log \frac{q(x)}{p(x)} \right]$,

- k_1 estimator: $\frac{1}{n} \sum_{i=1}^n \log \frac{q(x_i)}{p(x_i)}$. Unbiased, high variance. Can be negative.
- k_2 estimator: $\frac{1}{n} \sum_{i=1}^n \frac{1}{2} \left(\log \frac{p(x_i)}{q(x_i)} \right)^2$. biased, low variance. Always positive.
- k_3 estimator: $\frac{p(x)}{q(x)} - 1 - \log \frac{p(x)}{q(x)}$. Always positive. Better than k_2 ? (non-negativity: $\log a \leq a - 1$ for all $a > 0$).

vD: To add: comparison with three estimators

2 Policy Gradient

The goal of the policy-based RL is to maximize the expected reward

$$\max \left\{ \mathcal{J}(\theta) := \mathbb{E}_{\tau \sim \pi_\theta} [R(\tau)] = \int P(\tau|\theta) R(\tau) d\tau \right\}, \quad (5)$$

where $P(\tau|\theta)$ is defined in (2) and $s_0 \sim \rho_0(\cdot)$

- $a_t \sim \pi_\theta(\cdot|s_t)$ for all $t \in \{0, 1, \dots, T-1\}$.
- $s_{t+1} \sim P(\cdot|s_t, a_t)$ for all $t \in \{0, 1, \dots, T-1\}$.
- $r_t = R(s_t, a_t, s_{t+1})$ for all $t \in \{0, 1, \dots, T-1\}$.

Applying (2) and (3), we get

$$\begin{aligned} \nabla_\theta \mathcal{J}(\theta) &= \int \nabla_\theta P(\tau|\theta) R(\tau) d\tau = \int P(\tau|\theta) \nabla_\theta \log P(\tau|\theta) R(\tau) d\tau = \mathbb{E}_{\tau \sim \pi_\theta} [\nabla_\theta \log P(\tau|\theta) R(\tau)] \\ &= \mathbb{E}_{\tau \sim \pi_\theta} \left[\nabla_\theta \left(\rho_0(s_0) + \sum_{t=0}^{T-1} P(s_{t+1}|s_t, a_t) \right) + \sum_{t=0}^{T-1} \nabla_\theta \log \pi_\theta(a_t|s_t) \right] R(\tau) \\ &= \mathbb{E}_{\tau \sim \pi_\theta} \left[R(\tau) \sum_{t=0}^{T-1} \nabla_\theta \log \pi_\theta(a_t|s_t) \right] \end{aligned} \quad (6)$$

2.1 Policy Gradient Estimator

Given N trajectories $\{\tau_1, \dots, \tau_N\}$ with $\tau_n \sim \pi_\theta$ for all $n \in [N]$, how can we construct a "good" policy gradient estimator?

The natural choice would be

- Naive unbiased estimator

$$\widetilde{\nabla_\theta \mathcal{J}(\theta)} = \frac{1}{N} \sum_{n=1}^N \left(\sum_{t=0}^{T_n-1} R(\tau_n) \nabla_\theta \log \pi_\theta(a_t|s_t) \right)$$

However, the above looks a bit weird in the sense that $\log \pi_\theta(a_t|s_t)$ contains single-step probability while $R(\tau_n)$ represents the whole trajectory. This naturally leads to the following abstraction

$$\widetilde{\nabla_\theta \mathcal{J}(\theta)} = \frac{1}{N} \sum_{n=1}^N \left(\sum_{t=0}^{T_n-1} \Phi_t \nabla_\theta \log \pi_\theta(a_t|s_t) \right)$$

Some popular choices of Φ_t (Schulman et al., 2015), hence the gradient estimator can be

- Reward-to-go unbiased estimator (drops a_t 's dependency on the past actions)

$$\widetilde{\nabla_\theta \mathcal{J}(\theta)} = \frac{1}{N} \sum_{n=1}^N \left(\sum_{t=0}^{T_n-1} G_t(\tau_n) \nabla_\theta \log \pi_\theta(a_t|s_t) \right)$$

Recall $G_t(\tau) = \sum_{t'=t}^{T-1} \gamma^{t'-t} r_{t'}$. If one adopts this estimator, then the algorithm is called **REINFORCE** algorithm. This method drops the need on the Value function (see details below) but suffers from the high variance in gradient estimates.

- with baseline (variance reduction)

$$\widetilde{\nabla_\theta \mathcal{J}(\theta)} = \frac{1}{N} \sum_{n=1}^N \left(\sum_{t=0}^{T_n-1} (G_t(\tau_n) - b(s_t)) \nabla_\theta \log \pi_\theta(a_t|s_t) \right)$$

One popular choice of the baseline function is $b(s_t) := V^\pi(s_t)$. When V^π is unknown, we can approximate it with $V_\phi^\pi(\cdot)$, a neural network parameterized by ϕ .

- Advantages based estimator

$$\widetilde{\nabla_{\theta} \mathcal{J}(\theta)} = \frac{1}{N} \sum_{n=1}^N \left(\sum_{t=0}^{T_n-1} A^{\pi}(s_t, a_t) \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) \right),$$

where $A^{\pi}(s_t, a_t) = Q^{\pi}(s_t, a_t) - V^{\pi}(s_t)$.

- TD-error based estimator

$$\widetilde{\nabla_{\theta} \mathcal{J}(\theta)} = \frac{1}{N} \sum_{n=1}^N \left((r_t + \gamma V^{\pi}(s_{t+1}) - V^{\pi}(s_t)) \sum_{t=0}^{T_n-1} \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) \right),$$

where $r_t + \gamma V^{\pi}(s_{t+1}) - V^{\pi}(s_t)$ is a approximation of the $A^{\pi}(s_t, a_t)$.

- GAE(γ, λ) estimator

$$\widetilde{\nabla_{\theta} \mathcal{J}(\theta)} = \frac{1}{N} \sum_{n=1}^N \left(\sum_{\ell=0}^{T_n} (\gamma \lambda)^{\ell} \delta_{t+\ell} \sum_{t=0}^{T_n-1} \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) \right),$$

where $\delta_t = r_t + \gamma V_{\phi}^{\pi}(s_{t+1}) - V_{\phi}^{\pi}(s_t)$. GAE($\gamma, 0$) typically has low variance and GAE($\gamma, 1$) has high variance.

2.2 REINFORCE algorithm

If one adopts the Reward-to-go unbiased estimator, we can reach to the REINFORCE algorithm. The objective function is

$$\mathcal{J}(\theta) = \mathbb{E}_{\tau \sim \pi_{\theta}} \left(\sum_{t=0}^{T-1} G_t(\tau) \log \pi_{\theta}(a_t | s_t) \right) \quad (7)$$

where $G_t(\tau) = \sum_{t'=t}^{T-1} \gamma^{t'-t} r_{t'}$. This method drops the need on the Value function but suffers from the high variance in gradient estimates. The algorithm is summarized in Algorithm 1

Algorithm 1 REINFORCE

- 1: Inputs: Initial Policy network parameter θ_0 and stepsize α .
- 2: **for** $k=0,1,2,\dots$ **do**
- 3: Collect a set of trajectories $D_k = \{\tau_n\}_{n=1}^N$.
- 4: Reward-to-go calculation $G_t(\tau_n) = \sum_{t'=t}^{T_n-1} \gamma^{t'-t} r_{t'}$ for all n .
- 5: Compute the gradient estimate

$$\widetilde{\nabla_{\theta} \mathcal{J}(\theta_k)} = \frac{1}{N} \sum_{n=1}^N \left(\sum_{t=0}^{T_n-1} G_t(\tau_n) \nabla_{\theta} \log \pi_{\theta_k}(a_t | s_t) \right)$$

- 6: Policy update $\theta_{k+1} \leftarrow \theta_k + \alpha \widetilde{\nabla_{\theta} \mathcal{J}(\theta_k)}$
 - 7: **end for**
-

2.3 Actor-Critic Algorithm

When one adopts the estimator for the policy gradient, usually the state value function is used. Let us approximate it with V_{ϕ}^{π} the objective function for TD-residual based gradient estimator can written as

$$\max \left\{ \mathcal{J}_{\text{TD}}(\theta) := \mathbb{E}_{\tau \sim \pi_{\theta}} \left[\sum_{t=0}^{T-1} (r_t + \gamma V_{\phi}^{\pi}(s_{t+1}) - V_{\phi}^{\pi}(s_t)) \log \pi_{\theta}(a_t | s_t) \right] \right\}.$$

Similarly, the V_{ϕ}^{π} can be updated by

$$\min \left\{ \mathcal{L}(\phi) := \mathbb{E}_{\tau \sim \pi_{\theta}} \left[\sum_{t=0}^{T-1} [(r_t + \gamma V_{\phi}^{\pi}(s_{t+1}) - V_{\phi}^{\pi}(s_t))^2] \right] \right\}. \quad (8)$$

This is actually the actor-critic framework. When we drive $\mathcal{L}(\phi)$ to 0, intuitively, the advantage will be close to 0, which means that for the actor, π_{θ} is approaching the optimal policy and the value function is also approaching the optimal.

vd:To add

Algorithm 2 Naive Actor-Critic Algorithm

- 1: Inputs: Initial Policy network parameter θ_0 and stepsize α .
 - 2: **for** $k=0,1,2,\dots$ **do**
 - 3: Collect a set of trajectories $D_k = \{\tau_n\}_{n=1}^N$.
 - 4: Reward-to-go calculation $G_t(\tau_n) = \sum_{t'=t}^{T_n-1} \gamma^{t'-t} r'_{t'}$ for all n .
 - 5: Compute the gradient estimate
 - 6: Policy update $\theta_{k+1} \leftarrow \theta_k + \alpha \widetilde{\nabla_{\theta} \mathcal{J}(\theta_k)}$
 - 7: **end for**
-

Part II

Algorithms

3 Proximal Policy Optimization

3.1 PPO introduced by OpenAI

When performing the policy gradient update, each step one needs to sample the trajectories of the environment (since after one update on the policy π_θ , the trajectory becomes obsolete. How can we address the sample efficiency issue? Importance sampling comes to rescue.

Consider the gradient estimator function, where $\hat{A}_\phi$ is any "good" estimate of the advantage¹.

$$\begin{aligned}\widetilde{\nabla_\theta \mathcal{J}}(\theta) &= \mathbb{E}_{\tau \sim \pi_\theta} \left[\sum_{t=0}^{T-1} \hat{A}_\phi(s_t, a_t) \nabla_\theta \log \pi_\theta(a_t | s_t) \right] = \mathbb{E}_{\tau \sim \pi_{\theta_{\text{old}}}} \left[\sum_{t=0}^{T-1} \frac{\pi_\theta(a_t | s_t)}{\pi_{\theta_{\text{old}}}(a_t | s_t)} \hat{A}_\phi(s_t, a_t) \nabla_\theta \log \pi_\theta(a_t | s_t) \right] \\ &= \mathbb{E}_{\tau \sim \pi_{\theta_{\text{old}}}} \left[\sum_{t=0}^{T-1} \frac{1}{\pi_{\theta_{\text{old}}}(a_t | s_t)} \hat{A}_\phi(s_t, a_t) \nabla_\theta \pi_\theta(a_t | s_t) \right].\end{aligned}$$

The corresponding objective function can be written as

$$\mathcal{J}_{\text{advantage}}(\theta) := \mathbb{E}_{\tau \sim \pi_{\theta_{\text{old}}}} \left[\sum_{t=0}^{T-1} \frac{\pi_\theta(a_t | s_t)}{\pi_{\theta_{\text{old}}}(a_t | s_t)} \hat{A}_\phi(s_t, a_t) \right]$$

In practice, we cannot sample enough trajectories, hence the estimation of the $\mathcal{J}_{\text{advantage}}(\theta)$ through sample average is not accurate. This will be worsen by the fact that if $\pi_{\theta_{\text{old}}}$ is very different from π_θ . This leads to two versions of the PPO objective functions.

$$\max_{\theta} \mathcal{J}_{\text{PPO-clip}}(\theta) := \mathbb{E}_{\tau \sim \pi_{\theta_{\text{old}}}} \left[\sum_{t=0}^{T-1} \min \left\{ \frac{\pi_\theta(a_t | s_t)}{\pi_{\theta_{\text{old}}}(a_t | s_t)} \hat{A}_\phi(s_t, a_t), \text{clip} \left(\frac{\pi_\theta(a_t | s_t)}{\pi_{\theta_{\text{old}}}(a_t | s_t)}, 1 - \epsilon, 1 + \epsilon \right) \hat{A}_\phi(s_t, a_t) \right\} \right], \quad (9)$$

where

$$\text{clip}(x, 1 - \epsilon, 1 + \epsilon) = \begin{cases} x, & \text{if } 1 - \epsilon < x < 1 + \epsilon \\ 1 - \epsilon, & \text{if } x \leq 1 - \epsilon \\ 1 + \epsilon, & \text{if } x \geq 1 + \epsilon. \end{cases}$$

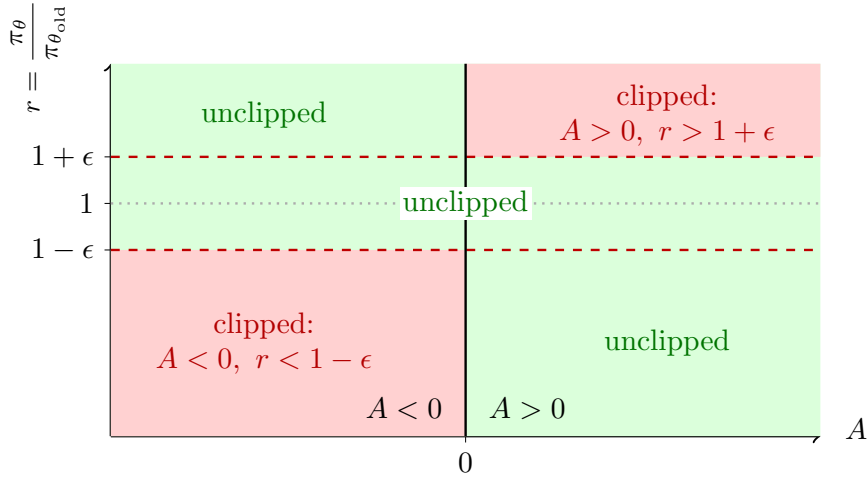


Figure 1: PPO-Clip trust region in the (A, r) plane, where $r = \pi_\theta(a|s)/\pi_{\theta_{\text{old}}}(a|s)$. The red region is the clipped region, and the green region is the unclipped region (after the clip and min operation).

Remark

- The popular choice of $\hat{A}_\phi(s_t, a_t)$ is $A_\phi^{\text{GAE}(\gamma, \lambda)} = \sum_{\ell=0}^{T-1-t} (\gamma\lambda)^\ell \left(r_{t+\ell} + \gamma V_\phi^\pi(s_{t+1+\ell}) - V_\phi^\pi(s_{t+\ell}) \right)$.

¹As the subscripts suggests, the definition of $\hat{A}_\phi$ usually rely on the value function V_ϕ^π .

- When $\hat{A}_\phi(s_t, a_t) > 0$, which means that a_t is better than the average, we want to increase its probability. However, we need to limit the increase to not being greater than the factor of $1 + \epsilon$.
- When $\hat{A}_\phi(s_t, a_t) < 0$, which means that a_t is worse than the average, we want to decrease its probability. However, we need to limit the decrease to be no greater than the factor of $1 - \epsilon$.

One can also constraint the distribution shift by

$$\max_{\theta} \mathcal{J}_{\text{PPO-KL}}(\theta) := \mathbb{E}_{\tau \sim \pi_{\theta_{\text{old}}}} \left[\sum_{t=0}^{T-1} \frac{\pi_{\theta}(a_t|s_t)}{\pi_{\theta_{\text{old}}}(a_t|s_t)} \hat{A}_{\phi}(s_t, a_t) - \beta \text{KL}[\pi_{\theta_{\text{old}}}(\cdot|s_t) || \pi_{\theta}(\cdot|s_t)] \right], \quad (10)$$

Remark

- When maximize $\mathcal{J}_{\text{PPO-KL}}(\theta)$, we minimize $\mathbb{D}_{\text{KL}}[\pi_{\theta_{\text{old}}}(\cdot|s_t) || \pi_{\theta}(\cdot|s_t)]$, i.e., we minimize the forward KL, which has the mean-seeking behavior.

To put pieces together, we summarize the PPO-Clip in Algorithm 3.

Algorithm 3 PPO-Clip Algorithm

- 1: Inputs: Initial Policy network parameter θ_0 and value function parameter ϕ_0 .
- 2: **for** $k=0,1,2,\dots$ **do**
- 3: Collect a set of trajectories $D_k = \{\tau_n\}_{n=1}^N$.
- 4: Reward-to-go calculation $G_t(\tau_n) = \sum_{t'=t}^{T_n-1} \gamma^{t'-t} r_{t'}$ for all n .
- 5: GAE calculation $A_{\phi_k}^{\text{GAE}(\gamma, \lambda)}(s_t^{(n)}, a_t^{(n)})$ for all t and n ².
- 6: Approximately solve

$$\theta_{k+1} = \arg \max_{\theta} \frac{1}{N} \sum_{n=1}^N \frac{1}{T_n} \sum_{t=0}^{T_n-1} \min \left\{ \frac{\pi_{\theta}(a_t^{(n)}|s_t^{(n)})}{\pi_{\theta_k}(a_t^{(n)}|s_t^{(n)})} A_{\phi_k}^{\text{GAE}(\gamma, \lambda)}(s_t^{(n)}, a_t^{(n)}), \right. \\ \left. \text{clip} \left(\frac{\pi_{\theta}(a_t|s_t)}{\pi_{\theta_k}(a_t|s_t)}, 1 - \epsilon, 1 + \epsilon \right) A_{\phi_k}^{\text{GAE}(\gamma, \lambda)}(s_t^{(n)}, a_t^{(n)}) \right\},$$

- 7: Approximately solve

$$\phi_{k+1} = \arg \min_{\phi} \frac{1}{N} \sum_{n=1}^N \frac{1}{T_n} \sum_{t=0}^{T_n-1} (V_{\phi}^{\pi}(s_t^{(n)}) - G_t(\tau_n))^2$$

- 8: **end for**
-

3.2 PPO in the context of RLHF

Let us first introduce a new set of notions and setup. Let $\mathcal{D}$ be a set of prompt and $x \in \mathcal{D}$ is any sample from it. $y \sim \pi_{\theta}(\cdot|x)$ be a response sampled from the policy model π_{θ} given the prompt x . To simplify the discussion, let us assume that the reward model is given and we focus on the outcome reward $r(x, y)$.

We consider the following variant objective function inspired by policy gradient and PPO

$$\mathcal{J}(\theta) = \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi_{\theta}(\cdot|x)} [r(x, y) - \beta \text{KL}[\pi_{\theta}(\cdot|x) || \pi_{\theta_{\text{ref}}}(\cdot|x)]] ,$$

note that here we used forward KL divergence. Then the gradient is

$$\nabla \mathcal{J}(\theta) = \int_{x \in \mathcal{X}} p(x) \left\{ \left[\int_{y \in \mathcal{Y}(x)} r(x, y) \frac{\partial}{\partial \theta} \pi_{\theta}(y|x) dy \right] - \beta \left[\int_{s \in \mathcal{Y}(x)} \frac{\partial}{\partial \theta} \pi_{\theta}(s|x) \log \frac{\pi_{\theta}(s|x)}{\pi_{\theta_{\text{ref}}}(s|x)} ds \right] \right\} dx, \quad (11)$$

where $\mathcal{X} = \mathcal{D}$ and $\mathcal{Y}(x) = \text{support}(\pi_{\theta}(y|x))$. Note that

$$\begin{aligned} & \int_{s \in \mathcal{Y}(x)} \frac{\partial}{\partial \theta} \pi_{\theta}(s|x) \log \frac{\pi_{\theta}(s|x)}{\pi_{\theta_{\text{ref}}}(s|x)} ds \\ &= \int_{s \in \mathcal{Y}(x)} \left[\nabla \pi_{\theta}(s|x) \cdot \log \frac{\pi_{\theta}(s|x)}{\pi_{\theta_{\text{ref}}}(s|x)} + \pi_{\theta}(s|x) \cdot \nabla \log \frac{\pi_{\theta}(s|x)}{\pi_{\theta_{\text{ref}}}(s|x)} \right] ds \\ &= \int_{s \in \mathcal{Y}(x)} \left[\nabla \pi_{\theta}(s|x) \cdot \log \frac{\pi_{\theta}(s|x)}{\pi_{\theta_{\text{ref}}}(s|x)} + \nabla \pi_{\theta}(s|x) \right] ds \\ &= \int_{s \in \mathcal{Y}(x)} \nabla \pi_{\theta}(s|x) \cdot \log \frac{\pi_{\theta}(s|x)}{\pi_{\theta_{\text{ref}}}(s|x)} ds \end{aligned} \quad (12)$$

where the last line follows from (4). Combining (11) and (12), one obtains

$$\begin{aligned}
\nabla \mathcal{J}(\theta) &= \int_{x \in \mathcal{X}} p(x) \left\{ \left[\int_{y \in \mathcal{Y}(x)} r(x, y) \frac{\partial}{\partial \theta} \pi_{\theta}(y|x) dy \right] - \beta \left[\int_{s \in \mathcal{Y}(x)} \nabla \pi_{\theta}(s|x) \cdot \log \frac{\pi_{\theta}(s|x)}{\pi_{\theta_{\text{ref}}}(s|x)} ds \right] \right\} dx \\
&= \int_{x \in \mathcal{X}, y \in \mathcal{Y}(x)} p(x) \left\{ \pi_{\theta}(y|x) \left[r(x, y) \nabla \log \pi_{\theta}(y|x) - \beta \nabla \log \pi_{\theta}(y|x) \cdot \log \frac{\pi_{\theta}(y|x)}{\pi_{\theta_{\text{ref}}}(y|x)} \right] \right\} dy dx \\
&= \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi_{\theta}(\cdot|x)} \left[\left(r(x, y) - \beta \log \frac{\pi_{\theta}(y|x)}{\pi_{\theta_{\text{ref}}}(y|x)} \right) \nabla \log \pi_{\theta}(y|x) \right] \tag{13}
\end{aligned}$$

From (13), we conclude the "PPO" version of the RLHF objective function

$$\max_{\theta} \mathcal{J}_{\text{RLHF}}(\theta) := \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi_{\theta}(\cdot|x)} \left[r(x, y) - \beta \log \frac{\pi_{\theta}(y|x)}{\pi_{\theta_{\text{ref}}}(y|x)} \right], \tag{14}$$

which recovers the first part of the objective function in Equation (2) of Ouyang et al. (2022).

4 REINFORCE++

REINFORCE++ modifies PPO in the LLM text by introducing

- token-level KL penalty $\text{KL}(t; \theta) = \log \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{\text{ref}}}(a_t|s_t)}$, where $\pi_{\theta_{\text{ref}}}$ can be the SFT model;
- token-level reward function $r_t = \mathbb{I}(s_t = [\text{EOS}])r(x, y) - \beta \text{KL}(t)$, where x and y are input prompt and response, respectively and $r(x, y)$ is the outcome reward.

Recall the advantage used in PPO is $A_\phi^{\text{GAE}(\gamma, \lambda)} = \sum_{\ell=0}^{T-1-t} (\gamma\lambda)^\ell (r_{t+\ell} + \gamma V_\phi^\pi(s_{t+1+\ell}) - V_\phi^\pi(s_{t+\ell}))$. If we drop the value network and plug in the new definition of r_t , we get

$$\begin{aligned} A_t^{\text{GAE}(\gamma, \lambda)} &= \sum_{\ell=0}^{T-1-t} (\gamma\lambda)^\ell [\mathbb{I}(s_{t+\ell} = [\text{EOS}])r(x, y) - \beta \text{KL}(t + \ell; \theta)] \\ &= \sum_{i=t}^{T-1} (\gamma\lambda)^\ell [\mathbb{I}(s_i = [\text{EOS}])r(x, y) - \beta \text{KL}(i)] = (\gamma\lambda)^{T-1} r(x, y) - \sum_{i=t}^{T-1} (\gamma\lambda)^i [\beta \text{KL}(i; \theta)]. \end{aligned}$$

Specifically, letting $\gamma = \lambda = 1$, we get the advantage estimation used in REINFORCE++ as

$$A_t^{\text{RF}++} = r(x, y) - \sum_{i=t}^{T-1} \beta \text{KL}(i; \theta). \quad (15)$$

Consequently, for the k th episode, REINFORCE++ solves the following problem

$$\theta_{k+1} = \arg \max_{\theta} \frac{1}{N} \sum_{n=1}^N \frac{1}{T_n} \sum_{t=0}^{T_n-1} \min \left\{ \frac{\pi_\theta(a_t^{(n)}|s_t^{(n)})}{\pi_{\theta_k}(a_t^{(n)}|s_t^{(n)})} \bar{A}_{n,t}, \text{clip} \left(\frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_k}(a_t|s_t)}, 1 - \epsilon, 1 + \epsilon \right) \bar{A}_{n,t} \right\},$$

where $\bar{A}_{n,t} = \frac{A_{n,t}^{\text{RF}++} - \mu_k}{\sigma_k}$, $\mu_k = \text{mean}\{A_{n,t}^{\text{RF}++}\}_{n \in [1, N], t \in [0, T_n-1]}$, $\sigma_k = \text{std}\{A_{n,t}^{\text{RF}++}\}_{n \in [1, N], t \in [0, T_n-1]}$, $A_{n,t}^{\text{RF}++} = r(x^{(n)}, y^{(n)}) - \sum_{i=t}^{T_n-1} \beta \text{KL}(i; \theta_k)$, and $(x^{(n)}, y^{(n)})$ is the n th sample.

To conclude, we summarize REINFORCE++ in Algorithm 4.

Algorithm 4 REINFORCE++

- 1: Inputs: Initial Policy network parameter θ_0 .
- 2: **for** $k=0, 1, 2, \dots$ **do**
- 3: Collect a set of trajectories $D_k = \{\tau_n\}_{n=1}^N$.
- 4: $\{A_{n,t}^{\text{RF}++}\}_{n \in [1, N], t \in [0, T_n-1]}$ calculation, batch mean μ_k and variance σ_k calculation.
- 5: Compute normalized advantage estimation $\{\bar{A}_{n,t} = \frac{A_{n,t}^{\text{RF}++} - \mu_k}{\sigma_k}\}_{n \in [1, N], t \in [0, T_n-1]}$.
- 6: Approximately solve

$$\theta_{k+1} = \arg \max_{\theta} \frac{1}{N} \sum_{n=1}^N \frac{1}{T_n} \sum_{t=0}^{T_n-1} \min \left\{ \frac{\pi_\theta(a_t^{(n)}|s_t^{(n)})}{\pi_{\theta_k}(a_t^{(n)}|s_t^{(n)})} \bar{A}_{n,t}, \text{clip} \left(\frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_k}(a_t|s_t)}, 1 - \epsilon, 1 + \epsilon \right) \bar{A}_{n,t} \right\},$$

7: **end for**

Remark

- Compared to GRPO, the normalization happens on the "batch" level; while GRPO applies the normalization within the "group" level.
- The KL penalty applied in the loss function is different. For GRPO, the KL penalty is taken out of the definition of the advantage estimation and uses the k_3 estimator (recall the definition in Definition 1.5), which is $\beta \mathbb{D}_{KL}[\pi_\theta || \pi_{\text{ref}}](i, t)$. In REINFORCE++, the KL penalty is factorized in the definition of the advantage estimation and uses the k_1 estimator, which is $\beta \sum_{i=t}^{T-1} \text{KL}(i; \theta)$. Notice that in the t th action, the penalty is the accumulation of all KL penalties starting from the time step t to the end.

Motivated by the discussion above, Xie et al. (2025) modifies the loss function by combining the GRPO and REINFORCE++ as below.

$$\theta_{k+1} = \arg \max_{\theta} \frac{1}{N} \sum_{n=1}^N \frac{1}{T_n} \sum_{t=0}^{T_n-1} \min \left\{ \frac{\pi_{\theta}(a_t^{(n)} | s_t^{(n)})}{\pi_{\theta_k}(a_t^{(n)} | s_t^{(n)})} \bar{A}_{n,t}, \text{clip} \left(\frac{\pi_{\theta}(a_t | s_t)}{\pi_{\theta_k}(a_t | s_t)}, 1 - \epsilon, 1 + \epsilon \right) \bar{A}_{n,t} - \beta \mathbb{D}_{KL} [\pi_{\theta} || \pi_{\text{ref}}] (i, t) \right\},$$

where $\bar{A}_{n,t} = \frac{A_{n,t}^{\text{Logic}} - \mu_k}{\sigma_k}$, $\mu_k = \text{mean}\{A_{n,t}^{\text{Logic}}\}_{n \in [1, N], t \in [0, T_n-1]}$, $\sigma_k = \text{std}\{A_{n,t}^{\text{Logic}}\}_{n \in [1, N], t \in [0, T_n-1]}$, $A_{n,t}^{\text{Logic}} = r(x^{(n)}, y^{(n)})$. Basically, it moves KL penalty out of the definition of the advantage estimators and use k_3 KL estimation.

5 Implicit Reward in Direct Preference Optimization

Direct Preference Optimization (DPO) does not explicitly train a separate reward model. Instead, it implicitly defines a reward function through the relationship between the optimized policy and the reference policy. DPO can be derived from the KL-regularized RLHF objective:

$$\max_{\pi} \mathbb{E}_{y \sim \pi(\cdot|x)} [r(x, y)] - \beta D_{\text{KL}}(\pi(\cdot|x) \parallel \pi_{\text{ref}}(\cdot|x)),$$

where π_{ref} is a fixed reference policy, usually the supervised fine-tuned model, and $\beta > 0$ controls the strength of the KL regularization.

The optimal policy of this objective satisfies

$$\pi^*(y|x) = \frac{1}{Z(x)} \pi_{\text{ref}}(y|x) \exp\left(\frac{1}{\beta} r(x, y)\right),$$

where $Z(x)$ is the partition function depending only on the prompt x .

Solving for the reward gives

$$r(x, y) = \beta \log \frac{\pi^*(y|x)}{\pi_{\text{ref}}(y|x)} + \beta \log Z(x).$$

Since $\beta \log Z(x)$ depends only on the prompt x , it cancels when comparing two responses to the same prompt. Therefore, DPO uses the following implicit reward parameterization:

$$r_{\theta}(x, y) = \beta \log \frac{\pi_{\theta}(y|x)}{\pi_{\text{ref}}(y|x)} + C(x),$$

where $C(x)$ is a prompt-dependent constant.

For a preferred response y_w and a rejected response y_l , the reward difference is

$$\begin{aligned} r_{\theta}(x, y_w) - r_{\theta}(x, y_l) &= \beta \left[\log \frac{\pi_{\theta}(y_w|x)}{\pi_{\text{ref}}(y_w|x)} - \log \frac{\pi_{\theta}(y_l|x)}{\pi_{\text{ref}}(y_l|x)} \right] \\ &= \beta [\log \pi_{\theta}(y_w|x) - \log \pi_{\theta}(y_l|x) - \log \pi_{\text{ref}}(y_w|x) + \log \pi_{\text{ref}}(y_l|x)]. \end{aligned}$$

Thus, the implicit reward in DPO measures how much the optimized policy increases the probability of a response relative to the reference policy.

5.1 DPO Objective

DPO is trained on preference data of the form

$$(x, y_w, y_l) \sim \mathcal{D},$$

where x is the prompt, y_w is the preferred response, and y_l is the rejected response.

The DPO loss is

$$\mathcal{L}_{\text{DPO}}(\theta) = -\mathbb{E}_{(x, y_w, y_l) \sim \mathcal{D}} \left[\log \sigma \left(\beta \left[\log \frac{\pi_{\theta}(y_w|x)}{\pi_{\text{ref}}(y_w|x)} - \log \frac{\pi_{\theta}(y_l|x)}{\pi_{\text{ref}}(y_l|x)} \right] \right) \right],$$

where $\sigma(\cdot)$ is the sigmoid function.

Define

$$\Delta_{\theta}(x, y_w, y_l) = \log \frac{\pi_{\theta}(y_w|x)}{\pi_{\text{ref}}(y_w|x)} - \log \frac{\pi_{\theta}(y_l|x)}{\pi_{\text{ref}}(y_l|x)}.$$

Then the DPO loss can be written compactly as

$$\mathcal{L}_{\text{DPO}}(\theta) = -\mathbb{E}_{(x, y_w, y_l) \sim \mathcal{D}} [\log \sigma(\beta \Delta_{\theta}(x, y_w, y_l))].$$

Minimizing this loss increases the relative log-probability of the preferred response over the rejected response, after correcting for their probabilities under the reference policy.

5.2 Reward Hacking in DPO

DPO does not use an explicit reward model during optimization. Therefore, it is less likely to suffer from the classical online reward hacking problem in PPO-style RLHF, where the policy repeatedly exploits weaknesses of a learned reward model.

However, DPO can still suffer from a different form of reward hacking or over-optimization. Since the implicit reward is determined by the preference dataset,

$$(x, y_w, y_l) \sim \mathcal{D},$$

DPO may exploit biases or artifacts in the preference data.

In this sense, DPO does not hack an explicit reward model. Instead, it may overfit or exploit the implicit preference signal encoded in the dataset:

DPO may hack the preference dataset rather than a learned reward model.

For example, if preferred responses are systematically longer than rejected responses, the model may learn the spurious rule

longer response $\Rightarrow$ better response.

If preferred responses often contain certain stylistic patterns, the model may learn to imitate those patterns even when they do not improve semantic quality.

Thus, DPO can still exhibit reward-hacking-like behavior through preference-data bias, distribution shift, or over-optimization of the implicit reward.

5.3 Common Failure Modes of DPO

A common failure mode of DPO is length bias. The sequence log-probability is

$$\log \pi_{\theta}(y \mid x) = \sum_{t=1}^{|y|} \log \pi_{\theta}(y_t \mid x, y_{<t}).$$

If preferred responses are usually longer than rejected responses, the model may learn to favor longer responses, even when they are not more accurate or helpful.

Another failure mode is style bias. If preferred responses share superficial stylistic features, such as excessive politeness, rigid formatting, or repeated templates, DPO may learn these features as proxies for quality.

DPO may also overfit to easy negatives. If the rejected response y_l is much worse than the preferred response y_w , the model only learns a coarse distinction:

$$y_w \gg y_l.$$

This may not teach the model how to distinguish between two high-quality responses.

Finally, DPO is an offline preference optimization method. It is trained on fixed preference data, but after training, the model generates responses from its own policy:

$$y \sim \pi_{\theta}(\cdot \mid x).$$

If this generation distribution differs significantly from the training preference data distribution, DPO may suffer from distribution shift.

5.4 Mitigating Reward Hacking and Over-Optimization in DPO

One important mitigation is improving the quality of the preference data. The preference dataset should minimize spurious correlations between preference labels and superficial features such as length, format, or style. Ideally, the preference label should reflect semantic quality rather than dataset artifacts.

To reduce length bias, one can use length-normalized log-probabilities. Instead of using

$$\log \pi_{\theta}(y \mid x) = \sum_{t=1}^{|y|} \log \pi_{\theta}(y_t \mid x, y_{<t}),$$

one may use the average log-probability

$$\frac{1}{|y|} \log \pi_{\theta}(y \mid x) = \frac{1}{|y|} \sum_{t=1}^{|y|} \log \pi_{\theta}(y_t \mid x, y_{<t}).$$

Another mitigation is tuning the temperature parameter β . The parameter β controls the strength of the preference margin in the DPO loss:

$$\log \sigma(\beta \Delta_{\theta}(x, y_w, y_l)).$$

If β is too large, the optimization may aggressively separate preferred and rejected responses, increasing the risk of over-optimization. A smaller β makes the update more conservative.

One can also monitor or explicitly regularize the divergence from the reference model:

$$D_{\text{KL}}(\pi_{\theta} \parallel \pi_{\text{ref}}).$$

Although DPO already uses the reference policy in its objective, explicit KL monitoring or early stopping can help prevent excessive drift from the reference model.

Another practical method is to mix the DPO loss with a supervised fine-tuning loss:

$$\mathcal{L}(\theta) = \mathcal{L}_{\text{DPO}}(\theta) + \lambda_{\text{SFT}} \mathcal{L}_{\text{SFT}}(\theta),$$

where

$$\mathcal{L}_{\text{SFT}}(\theta) = -\mathbb{E}_{(x, y_w) \sim \mathcal{D}} [\log \pi_{\theta}(y_w \mid x)].$$

The SFT loss helps preserve language quality, instruction-following behavior, and formatting stability.

Finally, iterative preference data collection can reduce distribution shift. Instead of training only on a fixed offline dataset, one can repeatedly sample responses from the current policy,

$$y \sim \pi_{\theta}(\cdot \mid x),$$

collect new preference labels, and continue preference optimization on data closer to the current model distribution.

6 Group Relative Policy Optimization

GRPO is motivated by removing the requirements on estimating the value function is used PPO Algorithm 3. For each question $q \sim P(Q)$, GRPO samples a group of outputs $\{o_1, o_2, \dots, o_G\}$ from the old policy $\pi_{\theta_{\text{old}}}$.

$$\mathcal{J}_{GRPO}(\theta) = \mathbb{E}_{[q \sim P(Q), \{o_i\}_{i=1}^G \sim \pi_{\theta_{\text{old}}}(O|q)]} \left[\frac{1}{G} \sum_{i=1}^G \frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \left\{ \min \left[\frac{\pi_{\theta}(o_{i,t}|q, o_{i,<t})}{\pi_{\theta_{\text{old}}}(o_{i,t}|q, o_{i,<t})} \hat{A}_{i,t}, \text{clip} \left(\frac{\pi_{\theta}(o_{i,t}|q, o_{i,<t})}{\pi_{\theta_{\text{old}}}(o_{i,t}|q, o_{i,<t})}, 1 - \varepsilon, 1 + \varepsilon \right) \hat{A}_{i,t} \right] - \beta \mathbb{D}_{KL}[\pi_{\theta}||\pi_{\text{ref}}](i, t) \right\} \right], \quad (16)$$

where

- $\mathbb{D}_{KL}[\pi_{\theta}||\pi_{\text{ref}}]$ is the k_3 estimator of the KL divergence with

$$\mathbb{D}_{KL}[\pi_{\theta}||\pi_{\text{ref}}](i, t) = \frac{\pi_{\text{ref}}(o_{i,t}|q, o_{i,<t})}{\pi_{\theta}(o_{i,t}|q, o_{i,<t})} - \log \frac{\pi_{\text{ref}}(o_{i,t}|q, o_{i,<t})}{\pi_{\theta}(o_{i,t}|q, o_{i,<t})} - 1$$

- the advantage estimate is defined as $\hat{A}_{i,t} := \frac{r_i - \text{mean}(\mathbf{r})}{\text{std}(\mathbf{r})}$ with $\mathbf{r} = \{r_1, r_2, \dots, r_G\}$.

Remark

- The reverse KL divergence is used, which has the mode-seeking behavior. The usage of k_3 estimator is crucial for GRPO because k_3 guarantee to be non-negative. Otherwise, if one adopts the k_1 estimator, then there could be a possibility to "hack" the training process by making the k_1 estimator arbitrarily small (very negative), then the reward is $\mathcal{J}_{GRPO}(\theta)$ is also maimized. However, this means that π_{θ} is very different from π_{ref} . At the same time, numerical instability is observed during the training dynamics. Because the computation of $\exp \left(\frac{\pi_{\text{ref}}(o_{i,t}|q, o_{i,<t})}{\pi_{\theta}(o_{i,t}|q, o_{i,<t})} \right)$ can be unstable.
- In the computation of the KL term, note that $o_i \sim \pi_{\theta_{\text{old}}}$. But the link suggests that o_i should be sampled from π_{θ} to make $\mathbb{D}_{KL}[\pi_{\theta}||\pi_{\text{ref}}]$ an unbiased estimator. In practice, to make these make sense, one has to initialize $\pi_{\theta} := \pi_{\theta_{\text{old}}}$ and take only one step update.
- When optimize the objective function, one of tries to $\min_{\theta} -\mathcal{J}_{GRPO}(\theta)$. One should notice that $-\mathcal{J}_{GRPO}$ can be smaller than 0, meaning the loss can be negative. This could happen when "min" term is larger than the KL term, i.e., when either the two distributions are quite different or the advantage is large.

7 Group Sequence Policy Optimization

In GRPO, the token-level importance sampling weight $\frac{\pi_\theta(o_{i,t}|q, o_{i,<t})}{\pi_{\theta_{\text{old}}}(o_{i,t}|q, o_{i,<t})}$ is applied.

GSPO argues that this is problematic since this weight is based on a single sample $o_{i,t}$ from each next-token distribution $\pi_{\theta_{\text{old}}}(o_{i,t}|q, o_{i,<t})$, it fails to perform the intended distribution-correction role³. Instead, it introduces high-variance noise into the training gradients, which accumulates over long sequences and is exacerbated by the clipping mechanism.

Therefore, GSPO proposes to use the sequence-level importance weight $\frac{\pi_\theta(o_i|q)}{\pi_{\theta_{\text{old}}}(o_i|q)}$. Therefore, the loss function becomes (ignoring the KL regularization term)

$$\mathcal{J}_{\text{GSPO}}(\theta) = \mathbb{E}_{[q \sim P(Q), \{o_i\}_{i=1}^G \sim \pi_{\theta_{\text{old}}}(O|q)]} \left[\frac{1}{G} \sum_{i=1}^G \min \left[s_i(\theta) \hat{A}_i, \text{clip}(s_i(\theta), 1 - \varepsilon, 1 + \varepsilon) \hat{A}_i \right] \right],$$

where $s_i(\theta) = \left(\frac{\pi_\theta(o_i|q)}{\pi_{\theta_{\text{old}}}(o_i|q)} \right)^{1/|o_i|} = \exp \left(\frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \log \left\{ \frac{\pi_\theta(o_{i,t}|q, o_{i,<t})}{\pi_{\theta_{\text{old}}}(o_{i,t}|q, o_{i,<t})} \right\} \right)$, and $\hat{A}_i = \frac{r(x, o_i) - \text{mean}(\{r(x, o_j)\}_{j=1}^G)}{\text{std}(\{r(x, o_j)\}_{j=1}^G)}$.

Therefore, GSPO applies clipping to entire responses instead of individual tokens to exclude the overly "off-policy" samples from gradient estimation, which matches both the sequence-level rewarding and optimization. Also, note that the clipping ranges in GSPO and in previous algorithms (e.g., GRPO) typically differ in order of magnitude due to the distinct definitions of importance ratios.

In scenarios like multi-turn RL, we may desire a finer-grained advantage adjustment than the sequence level.

$$\mathcal{J}_{\text{GSPO-token}}(\theta) = \mathbb{E}_{[q \sim P(Q), \{o_i\}_{i=1}^G \sim \pi_{\theta_{\text{old}}}(O|q)]} \left[\frac{1}{G} \sum_{i=1}^G \frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \left\{ \min \left[s_{i,t}(\theta) \hat{A}_{i,t}, \text{clip}(s_{i,t}(\theta), 1 - \varepsilon, 1 + \varepsilon) \hat{A}_{i,t} \right] \right\} \right],$$

where $s_{i,t}(\theta) = \text{sg}[s_i(\theta)] \frac{\pi_\theta(o_{i,t}|q, o_{i,<t})}{\text{sg}[\pi_\theta(o_{i,t}|q, o_{i,<t})]}$ and sg is the stop gradient operator, so numerically, $s_{i,t}(\theta) = s_i(\theta)$.

To see more clearly about the gradient of the GSPO, GSPO-token, and GRPO, we can compute (clipping is omitted for brevity)

$$\begin{aligned} \nabla_\theta \mathcal{J}_{\text{GSPO}}(\theta) &= \nabla_\theta \mathbb{E}_{x \sim \mathcal{D}, \{o_i\}_{i=1}^G \sim \pi_{\theta_{\text{old}}}(\cdot|x)} \left[\frac{1}{G} \sum_{i=1}^G s_i(\theta) \hat{A}_i \right] \\ &= \mathbb{E}_{x \sim \mathcal{D}, \{o_i\}_{i=1}^G \sim \pi_{\theta_{\text{old}}}(\cdot|x)} \left[\frac{1}{G} \sum_{i=1}^G s_i(\theta) \hat{A}_i \cdot \nabla_\theta \log s_i(\theta) \right] \\ &= \mathbb{E}_{x \sim \mathcal{D}, \{o_i\}_{i=1}^G \sim \pi_{\theta_{\text{old}}}(\cdot|x)} \left[\frac{1}{G} \sum_{i=1}^G \left(\frac{\pi_\theta(o_i|x)}{\pi_{\theta_{\text{old}}}(o_i|x)} \right)^{\frac{1}{|o_i|}} \hat{A}_i \cdot \frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \nabla_\theta \log \pi_\theta(o_{i,t} | x, o_{i,<t}) \right] \\ \nabla_\theta \mathcal{J}_{\text{GSPO-token}}(\theta) &= \nabla_\theta \mathbb{E}_{x \sim \mathcal{D}, \{o_i\}_{i=1}^G \sim \pi_{\theta_{\text{old}}}(\cdot|x)} \left[\frac{1}{G} \sum_{i=1}^G \frac{1}{|o_i|} \sum_{t=1}^{|o_i|} s_{i,t}(\theta) \hat{A}_{i,t} \right] \\ &= \mathbb{E}_{x \sim \mathcal{D}, \{o_i\}_{i=1}^G \sim \pi_{\theta_{\text{old}}}(\cdot|x)} \left[\frac{1}{G} \sum_{i=1}^G s_i(\theta) \cdot \frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \hat{A}_{i,t} \frac{\nabla_\theta \pi_\theta(o_{i,t} | x, o_{i,<t})}{\pi_\theta(o_{i,t} | x, o_{i,<t})} \right] \\ &= \mathbb{E}_{x \sim \mathcal{D}, \{o_i\}_{i=1}^G \sim \pi_{\theta_{\text{old}}}(\cdot|x)} \left[\frac{1}{G} \sum_{i=1}^G \left(\frac{\pi_\theta(o_i|x)}{\pi_{\theta_{\text{old}}}(o_i|x)} \right)^{\frac{1}{|o_i|}} \cdot \frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \hat{A}_{i,t} \nabla_\theta \log \pi_\theta(o_{i,t} | x, o_{i,<t}) \right] \\ \nabla_\theta \mathcal{J}_{\text{GRPO}}(\theta) &= \nabla_\theta \mathbb{E}_{x \sim \mathcal{D}, \{o_i\}_{i=1}^G \sim \pi_{\theta_{\text{old}}}(\cdot|x)} \left[\frac{1}{G} \sum_{i=1}^G \frac{1}{|o_i|} \sum_{t=1}^{|o_i|} w_{i,t}(\theta) \hat{A}_{i,t} \right] \\ &= \mathbb{E}_{x \sim \mathcal{D}, \{o_i\}_{i=1}^G \sim \pi_{\theta_{\text{old}}}(\cdot|x)} \left[\frac{1}{G} \sum_{i=1}^G \hat{A}_i \cdot \frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \frac{\pi_\theta(o_{i,t} | x, o_{i,<t})}{\pi_{\theta_{\text{old}}}(o_{i,t} | x, o_{i,<t})} \nabla_\theta \log \pi_\theta(o_{i,t} | x, o_{i,<t}) \right] \end{aligned}$$

³Recall that Importance sampling needs multiple samples from the same distribution to reduce variance.

Therefore, the fundamental distinction between GSPO and GRPO lies in how they weight the gradients of the log likelihoods of tokens. In GRPO, the tokens are weighted according to their respective "importance weight". In contrast, GSPO weights all the tokens in a response equally, eliminating this instability factor of GRPO. GSPO-token and GSPO are numerically identical in the optimization objective, clipping condition, and theoretical gradient when we set the advantages of all the tokens in the response o_i to the same value (i.e., $\hat{A}_{i,t} = \hat{A}_i$), while GSPO-token enjoys the higher flexibility of adjusting the advantages per token.

$$r_{i,t} = \begin{cases} r_{i,t}^{\text{re}} + \gamma^{T_i-t} R_{\text{out},i}, & R_{\text{out},i} = +1, \\ \text{sim}_{i^*,t} \cdot r_{i^*,t}^{\text{re}} + \gamma^{T_i-t} R_{\text{out},i}, & R_{\text{out},i} = -1. \end{cases} \quad (17)$$

8 Clipped Importance Sampling Policy Optimization

Clipped Importance Sampling Policy Optimization (CISPO (Chen et al., 2025)) proposed in MiniMax-M1 optimizes a policy-gradient objective using a clipped importance sampling ratio as a stop-gradient weight. The CISPO objective is defined as

$$\mathcal{J}_{\text{CISPO}}(\theta) = \mathbb{E}_{(q,a) \sim \mathcal{D}, \{o_i\}_{i=1}^G \sim \pi_{\theta_{\text{old}}}(\cdot|q)} \left[\frac{1}{\sum_{i=1}^G |o_i|} \sum_{i=1}^G \sum_{t=1}^{|o_i|} \text{sg}(\hat{r}_{i,t}(\theta)) \hat{A}_{i,t} \log \pi_{\theta}(o_{i,t} | q, o_{i,<t}) \right].$$

Here, $\hat{A}_{i,t}$ is the estimated advantage for token $o_{i,t}$, and $\text{sg}(\cdot)$ denotes the stop-gradient operation. Therefore, $\text{sg}(\hat{r}_{i,t}(\theta))$ is treated as a constant when taking gradients with respect to θ . The clipped importance sampling ratio is

$$\hat{r}_{i,t}(\theta) = \text{clip}(r_{i,t}(\theta), 1 - \epsilon_{\text{low}}, 1 + \epsilon_{\text{high}}),$$

where the token-level importance sampling ratio is

$$r_{i,t}(\theta) = \frac{\pi_{\theta}(o_{i,t} | q, o_{i,<t})}{\pi_{\theta_{\text{old}}}(o_{i,t} | q, o_{i,<t})}.$$

Equivalently,

$$\hat{r}_{i,t}(\theta) = \begin{cases} 1 - \epsilon_{\text{low}}, & r_{i,t}(\theta) < 1 - \epsilon_{\text{low}}, \\ r_{i,t}(\theta), & 1 - \epsilon_{\text{low}} \leq r_{i,t}(\theta) \leq 1 + \epsilon_{\text{high}}, \\ 1 + \epsilon_{\text{high}}, & r_{i,t}(\theta) > 1 + \epsilon_{\text{high}}. \end{cases}$$

Thus, CISPO does not clip the surrogate objective itself. Instead, it clips the token-level importance sampling ratio and uses the clipped ratio as a stop-gradient coefficient in front of the log-probability term.

8.1 Gradient Difference between PPO and CISPO

The main difference between PPO and CISPO is how clipping affects the policy gradient. For PPO, the clipped surrogate objective at token (i, t) can be written as

$$L_{i,t}^{\text{PPO}}(\theta) = \min \left(r_{i,t}(\theta) \hat{A}_{i,t}, \text{clip}(r_{i,t}(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_{i,t} \right),$$

where

$$r_{i,t}(\theta) = \frac{\pi_{\theta}(o_{i,t} | q, o_{i,<t})}{\pi_{\theta_{\text{old}}}(o_{i,t} | q, o_{i,<t})}.$$

Consider the case

$$\hat{A}_{i,t} > 0 \quad \text{and} \quad r_{i,t}(\theta) > 1 + \epsilon.$$

Then the clipped PPO objective becomes

$$L_{i,t}^{\text{PPO}}(\theta) = (1 + \epsilon) \hat{A}_{i,t}.$$

This term is constant with respect to θ , and therefore

$$\nabla_{\theta} L_{i,t}^{\text{PPO}}(\theta) = 0.$$

Thus, when the probability of a positive-advantage token has already increased beyond the clipping range, PPO removes the gradient contribution from this token.

Similarly, if

$$\hat{A}_{i,t} < 0 \quad \text{and} \quad r_{i,t}(\theta) < 1 - \epsilon,$$

then the clipped PPO objective becomes

$$L_{i,t}^{\text{PPO}}(\theta) = (1 - \epsilon) \hat{A}_{i,t},$$

which is also constant with respect to θ . Hence,

$$\nabla_{\theta} L_{i,t}^{\text{PPO}}(\theta) = 0.$$

In this case, PPO removes the gradient contribution when the probability of a negative-advantage token has already decreased beyond the clipping range.

By contrast, CISPO uses the clipped importance sampling ratio as a stop-gradient coefficient:

$$\mathcal{J}_{\text{CISPO}}(\theta) = \mathbb{E} \left[\frac{1}{\sum_{i=1}^G |o_i|} \sum_{i=1}^G \sum_{t=1}^{|o_i|} \text{sg}(\hat{r}_{i,t}(\theta)) \hat{A}_{i,t} \log \pi_{\theta}(o_{i,t} \mid q, o_{i,<t}) \right].$$

Since

$$\text{sg}(\hat{r}_{i,t}(\theta))$$

is treated as a constant during backpropagation, the gradient of the CISPO objective is

$$\nabla_{\theta} \mathcal{J}_{\text{CISPO}}(\theta) = \mathbb{E} \left[\frac{1}{\sum_{i=1}^G |o_i|} \sum_{i=1}^G \sum_{t=1}^{|o_i|} \text{sg}(\hat{r}_{i,t}(\theta)) \hat{A}_{i,t} \nabla_{\theta} \log \pi_{\theta}(o_{i,t} \mid q, o_{i,<t}) \right].$$

Therefore, even when the ratio $r_{i,t}(\theta)$ falls outside the clipping interval, CISPO does not make the gradient of that token vanish. Instead, the clipped ratio $\hat{r}_{i,t}(\theta)$ controls the magnitude of the gradient, while the token still contributes a policy-gradient signal through

$$\nabla_{\theta} \log \pi_{\theta}(o_{i,t} \mid q, o_{i,<t}).$$

In short, PPO clips the surrogate objective, which can cause clipped tokens to have zero gradient. CISPO clips the importance sampling ratio and uses it as a stop-gradient weight, so clipped tokens can still contribute to the policy gradient.

9 Context Compression

Part III

Asynchronous RL

For agentic workloads, especially long-horizon tasks, rollout lengths are highly variable, so short trajectories finish quickly while long ones become stragglers.

Asynchronous RL algorithms are proposed to mitigate this imbalanced generation overhead by consuming rollouts continuously as they arrive, improving utilization and wall-clock efficiency. However, asynchrony introduces one challenge, where each trajectory can be generated by multiple versions of the old rollout model, which leads to more unpredictable and severe off-policy, and thus harms the training stability.

Recall that in the Standard PPO, the objective is defined as

$$\max_{\theta} \mathcal{J}_{\text{PPO-clip}}(\theta) := \mathbb{E}_{\tau \sim \pi_{\theta_{\text{old}}}} \left[\sum_{t=0}^{T-1} \min \left\{ \frac{\pi_{\theta}(a_t|s_t)}{\pi_{\theta_{\text{old}}}(a_t|s_t)} \hat{A}_{\phi}(s_t, a_t), \text{clip} \left(\frac{\pi_{\theta}(a_t|s_t)}{\pi_{\theta_{\text{old}}}(a_t|s_t)}, 1 - \epsilon, 1 + \epsilon \right) \hat{A}_{\phi}(s_t, a_t) \right\} \right],$$

controls policy update size via the ratio $\frac{\pi_{\theta}}{\pi_{\theta_{\text{old}}}}$, where $\pi_{\theta_{\text{old}}}$ is assumed to be both the *proximal policy* (the anchor policy for PPO clipping) and the *behavior policy* (the policy that collected the data); both terms are defined in (Hilton et al., 2022), which was primarily proposed to achieve batch size-invariance for PPO.

The key idea is to decouple these two roles. Let $\pi_{\theta_{\text{prox}}}$ denote the proximal policy that anchors the clipping region, and let $\pi_{\theta_{\text{beh}}}$ denote the behavior policy that collected the trajectories. Then a three-policy version of PPO-Clip can be written as

$$\max_{\theta} \mathcal{J}_{\text{PPO-Decoupled}}(\theta) := \mathbb{E}_{\tau \sim \pi_{\theta_{\text{beh}}}} \left[\sum_{t=0}^{T-1} w_t \min \left\{ r_t(\theta) \hat{A}_{\phi}(s_t, a_t), \text{clip} \left(r_t(\theta), 1 - \epsilon, 1 + \epsilon \right) \hat{A}_{\phi}(s_t, a_t) \right\} \right],$$

where

$$w_t := \frac{\pi_{\theta_{\text{prox}}}(a_t|s_t)}{\pi_{\theta_{\text{beh}}}(a_t|s_t)}, \quad r_t(\theta) := \frac{\pi_{\theta}(a_t|s_t)}{\pi_{\theta_{\text{prox}}}(a_t|s_t)}.$$

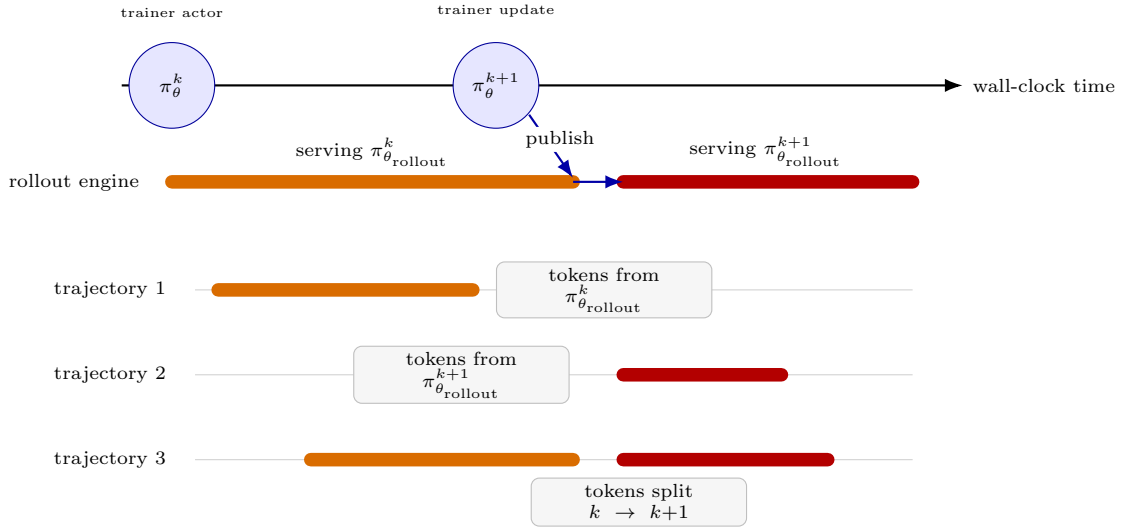
Here, w_t is the importance sampling correction from the behavior policy to the proximal policy. During one policy update, $\pi_{\theta_{\text{prox}}}$ and $\pi_{\theta_{\text{beh}}}$ are fixed, so w_t is treated as a constant with respect to θ . In contrast, $r_t(\theta)$ is the PPO ratio used to control how far the current policy π_{θ} moves away from the proximal policy $\pi_{\theta_{\text{prox}}}$.

Remark

- If $\pi_{\theta_{\text{beh}}} = \pi_{\theta_{\text{prox}}} = \pi_{\theta_{\text{old}}}$, then $w_t = 1$ and the objective reduces to the standard PPO-Clip objective.
- The behavior policy $\pi_{\theta_{\text{beh}}}$ can be stale or generated by a different worker, while the update size is still controlled relative to $\pi_{\theta_{\text{prox}}}$.
- The clipping mechanism is preserved because the clipped ratio is $r_t(\theta) = \pi_{\theta}(a_t|s_t)/\pi_{\theta_{\text{prox}}}(a_t|s_t)$, not the behavior correction w_t .

For the asynchronous discussion, it is clearer to write the proximal policy as $\pi_{\theta_{\text{old}}}$ and the behavior policy as $\pi_{\theta_{\text{rollout}}}$. The policy $\pi_{\theta_{\text{old}}}$ is the frozen PPO anchor used in the clipping ratio, while $\pi_{\theta_{\text{rollout}}}$ is the model that actually generated the sampled token or action. For more details, see the veRL discussion and the Slime reference implementation.

Consequently, the data consumed by a trainer update may not have a single behavior policy. Some trajectories may have been generated before the rollout engine refreshed, while later trajectories may use the refreshed rollout weights; in a long rollout, even different tokens in the same trajectory can correspond to different rollout snapshots. See Figure 2 for a visualization.



A training window can contain rollouts from different rollout-engine snapshots.

The stored rollout log-probability identifies the behavior policy for each sampled token.

Figure 2: Policy lag comes from the time gap between the trainer actor update and the rollout engine refresh. During this gap, the trainer has advanced to π_{θ}^{k+1} while the rollout engine may still generate tokens with $\pi_{\theta_{\text{rollout}}}^k$; after refresh, later tokens are generated with $\pi_{\theta_{\text{rollout}}}^{k+1}$. A trajectory that spans the refresh can therefore contain tokens from both rollout snapshots.

Thus, exact off-policy correction requires the behavior probability for the particular sampled token/action, namely the rollout-time value $\pi_{\theta_{\text{rollout}}}(a_t|s_t)$ used when that token/action was sampled. In practice, this means storing the rollout log-probability together with each generated token. If these log-probabilities are not stored, then one would need to keep enough rollout-policy checkpoints to recompute them later. The required history is therefore a history of behavior probabilities, or equivalently rollout-policy snapshots. In the long horizon setting, maintaining the history is not feasible in practice.

In this part, we discuss various asynchronous RL algorithms target to address various issues arised in the asynchronous RL.

10 SAO

10.1 Algorithm

To address the policy lag issue, SAO (Hou et al., 2026) proposes several modifications over the PPO-Clip objective to reach the following objective:

$$\max_{\theta} \mathcal{J}_{\text{SAO}}(\theta) := \mathbb{E}_{\tau \sim \pi_{\theta_{\text{rollout}}}} \left\{ \sum_{t=0}^{T-1} \left[\text{clip} \left(\frac{\pi_{\theta}(a_t | s_t)}{\pi_{\theta_{\text{rollout}}}(a_t | s_t)}, 1 - \epsilon_{\text{low}}, 1 + \epsilon_{\text{high}} \right) \hat{A}_{\phi}(s_t, a_t) \right] \log \pi_{\theta}(a_t | s_t) \right\}, \quad (18)$$

where (18) is written in REINFORCE form (7): the clipped importance ratio times the advantage is treated as a detached weight, and the explicit $\log \pi_{\theta}(a_t | s_t)$ term provides the policy-gradient direction.

Remark

1. SAO drops the proximal policy $\pi_{\theta_{\text{old}}}$ and uses the behavior policy $\pi_{\theta_{\text{rollout}}}$ for importance sampling:

$$r_t(\theta) = \frac{\pi_{\theta}(a_t | s_t)}{\pi_{\theta_{\text{rollout}}}(a_t | s_t)}.$$

This also avoids recomputing probabilities under $\pi_{\theta_{\text{old}}}$ by reusing the log-probabilities generated during rollout.

2. SAO applies a different trust region by clipping the rollout ratio to $[1 - \epsilon_{\text{low}}, 1 + \epsilon_{\text{high}}]$. As shown in Figure 3, PPO clips by asking “Has this sample already moved too far in the direction that appears beneficial?”, while SAO clips by asking “Is the current policy still sufficiently close to the policy that generated this token for this sample to be trustworthy?”.
- **PPO’s asymmetric construction is useful** when the advantage estimation is reliable, the data is fresh, the large ratio arose from parameter coupling or prior minibatch updates, and we want the sample to help undo an undesirable policy shift. For example, if $A = -1$ and $r = 10$, PPO-Clip keeps the directionally corrective term $rA = -10$, which pushes the sampled action’s probability down, while SAO clips it. Similarly, if $A = 1$ and $r = 0.1$, PPO keeps the corrective gradient, while SAO clips it. Because only one side is saturated for each advantage sign, PPO generally uses more of the batch and supports repeated minibatch optimization over collected data. However, PPO’s retained corrective gradients can become dangerous in asynchronous or heavily off-policy training.
 - **SAO’s symmetric construction is helpful** in asynchronous or heavily off-policy training. For $A < 0$ and $r \gg 1$, the factor rA can dominate a minibatch. That may be acceptable if A is accurate and the sample is still relevant, but it can be destructive when the sample was generated by a substantially older actor, the trajectory context is highly off-policy, or one extreme token dominates the aggregate gradient. SAO bounds the importance ratio regardless of the advantage sign, giving direct control over this source of gradient heavy tails.

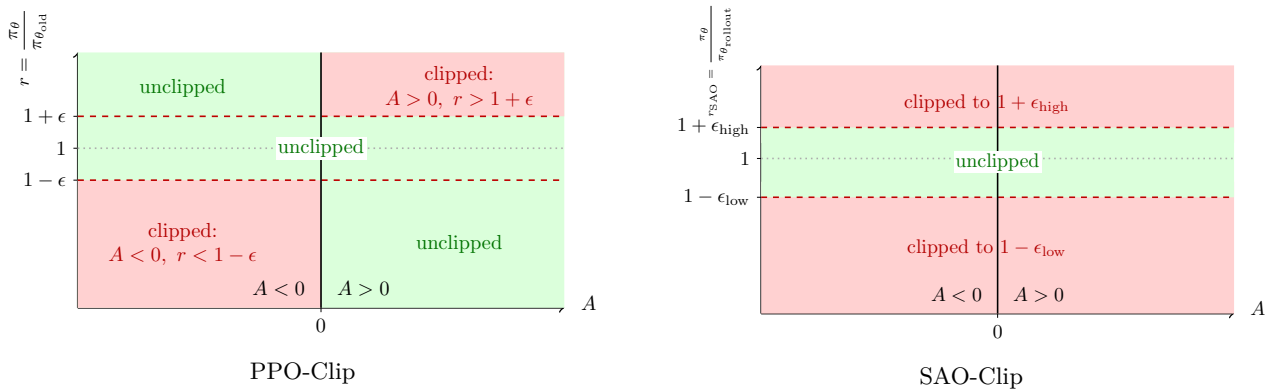


Figure 3: PPO-Clip and SAO-Clip trust regions in the (A, r) plane. PPO clips only in the regions selected by the clipped surrogate objective, while SAO directly clips the rollout-ratio $r = \frac{\pi_{\theta}(a_t | s_t)}{\pi_{\theta_{\text{rollout}}}(a_t | s_t)}$ to $[1 - \epsilon_{\text{low}}, 1 + \epsilon_{\text{high}}]$.

3. The strict two-sided gate has costs. First, once a token is outside the band, it gives no direct gradient, even when its advantage would have produced a restoring update. This is a distinction between filtering and constraining: SAO filters data; it does not project or roll the policy back into a feasible region. Second,

importance sampling is unbiased only without truncation under the usual assumptions. PPO introduces bias through one-sided clipping, while SAO introduces more bias by dropping both tails. In particular, dropping a negative-advantage outlier can make the observed surrogate look artificially favorable because a large negative contribution has disappeared. Therefore, SAO’s gated objective does not have PPO’s uniform pointwise pessimism property. Third, clipping reduces the effective batch size: if many tokens leave the band, the nominal batch size may remain large while the number of contributing tokens collapses.

4. The paper does not specify every detail of the advantage estimator $\hat{A}_\phi$. We sketch a consistent token-level interpretation below.

10.2 Guessed Advantage Estimator

Trajectory, action spans, and states. Let P denote the fixed task context supplied before interaction begins. It may contain system instructions, user instructions, task metadata, and chat-template tokens preceding the environment state. Before the agent takes its first action, the environment provides an initial observation o_0 . The interaction then proceeds as

$$o_0 \xrightarrow{a_0, R_0} o_1 \xrightarrow{a_1, R_1} o_2 \cdots \xrightarrow{a_{M-1}, R_{M-1}} o_M.$$

Equivalently, the chronological trace is

$$P, o_0, a_0, o_1, a_1, \dots, o_{M-1}, a_{M-1}, o_M.$$

Here o_i is the observation available before action a_i ; a_i is the i -th model-generated action span; after executing a_i , the environment returns reward R_i and next observation o_{i+1} ; and o_M is the final observation after the last sampled action. Action i consists of n_i policy-generated tokens:

$$a_i = (a_{i,0}, a_{i,1}, \dots, a_{i,n_i-1}).$$

These are exactly the tokens sampled from the policy during the i -th assistant generation. Tokens inserted by the chat template, such as an assistant-role marker, are not included in a_i unless they are actually sampled by the policy. Similarly, observation tokens are not part of a_i because they are supplied by the environment rather than generated by the policy.

Let σ_i denote the serialized assistant prefix placed immediately before generation of action a_i . Depending on the chat template, this may include tokens such as `<assistant>`, or tool-call formatting tokens. Define c_i as the complete token sequence available immediately before the first generated token $a_{i,0}$. Concretely,

$$c_i := \text{Serialize}(P, o_0, a_0, o_1, a_1, \dots, a_{i-1}, o_i, \sigma_i) \text{ for } i \geq 0.$$

For the j -th token of action span i , define $a_{i,<j} := (a_{i,0}, \dots, a_{i,j-1})$. The policy-decision state before generating token $a_{i,j}$ is $s_{i,j} := c_i \parallel a_{i,<j}$, where $\parallel$ denotes token-sequence concatenation. Thus $s_{i,0} = c_i$. The token-generation rule is

$$a_{i,j} \sim \pi_\theta(\cdot \mid s_{i,j}),$$

and the value associated with the same pre-token state is

$$v_{i,j} := V_\phi(s_{i,j}).$$

This notation keeps the actor and critic aligned: both **evaluate the prefix before the token is sampled**.

Successor states. Inside action span i , for $j < n_i - 1$, the next decision state is

$$s_{i,j+1} = s_{i,j} \parallel a_{i,j}.$$

At the end of the action span, the agent emits a_{i,n_i-1} , the environment executes the completed action, and returns o_{i+1} . The next policy-decision state is

$$s_{i+1,0} = c_{i+1}.$$

Expanded,

$$s_{i+1,0} = \text{Serialize}(P, o_0, a_0, o_1, \dots, o_i, a_i, o_{i+1}, \sigma_{i+1}).$$

The complete action a_i , not merely its final token, determines the environment transition. The final token is simply the token-level decision to which the action-level reward R_i is conventionally aligned. Observation o_{i+1} is part of the next decision state, but individual observation tokens are not separate policy-decision steps.

For two action spans, the causal structure is

$$P, o_0, \sigma_0 \xrightarrow{\text{generate } a_0} a_0 \xrightarrow{\text{environment}} R_0, o_1, \sigma_1 \xrightarrow{\text{generate } a_1} a_1 \xrightarrow{\text{environment}} R_1, o_2.$$

The corresponding policy-decision sequence is

$$s_{0,0}, s_{0,1}, \dots, s_{0,n_0-1}, s_{1,0}, s_{1,1}, \dots,$$

where

$$s_{0,0} = \text{Serialize}(P, o_0, \sigma_0), \quad s_{1,0} = \text{Serialize}(P, o_0, a_0, o_1, \sigma_1).$$

This is the sequence on which skip-observation GAE should operate. The observation o_1 is part of $s_{1,0}$, but its individual serialized tokens are not separate policy-decision steps.

Transformer and value-head alignment. Let

$$x_0, x_1, \dots, x_{T-1}$$

be the complete serialized token sequence, including prompt tokens, observations, action tokens, role tokens, tool delimiters, and assistant-prefix tokens. Let $p_{i,j}$ be the physical position of action token $a_{i,j}$, so

$$x_{p_{i,j}} = a_{i,j}.$$

By the state definition,

$$s_{i,j} = x_{<p_{i,j}}.$$

Write the value model as a causal transformer body followed by a scalar head:

$$h_t^V = F_\phi^{\text{body}}(x_{\leq t}), \quad \bar{v}_t = g_\phi^{\text{head}}(h_t^V),$$

with $\phi = (\phi_{\text{body}}, \phi_{\text{head}})$. The raw output $\bar{v}_t$ evaluates the prefix through physical position t :

$$\bar{v}_t = V_\phi(x_{\leq t}).$$

Since the state before action token $a_{i,j}$ is

$$s_{i,j} = x_{<p_{i,j}} = x_{\leq p_{i,j}-1},$$

its value is

$$v_{i,j} = V_\phi(s_{i,j}) = \bar{v}_{p_{i,j}-1}.$$

This means:

- before an interior action token, the relevant hidden state is usually located at the preceding action token;
- before the first token of a new action span, the relevant hidden state is located at the final serialized token before generation begins;
- that final preceding token may be an observation token, tool delimiter, or assistant-role token.

For the first token after observation o_{i+1} ,

$$v_{i+1,0} = V_\phi(s_{i+1,0}) = \bar{v}_{p_{i+1,0}-1}.$$

Because $s_{i+1,0} = x_{<p_{i+1,0}}$ contains o_{i+1} , the critic prediction $v_{i+1,0}$ is conditioned on the complete next observation.

Reward alignment. Let $r_{i,j}$ be the reward assigned to the transition caused by choosing token $a_{i,j}$. Inside an action span, the usual assignment is

$$r_{i,j} = 0, \quad j < n_i - 1,$$

unless token-level shaping rewards are used. If the environment returns reward R_i after executing the complete action span a_i , assign it to the span's final token:

$$r_{i,n_i-1} = R_i.$$

Equivalently,

$$r_{i,j} = \begin{cases} 0, & j < n_i - 1, \\ R_i, & j = n_i - 1. \end{cases}$$

For a trajectory with only one terminal verifier reward R , all token rewards are zero except the final token of the final action span. This does not claim that the final token alone caused the reward; earlier action tokens receive credit through the backward GAE recursion.

Token-level TD residuals and advantages. For an interior token $j < n_i - 1$, the next policy-decision state is $s_{i,j+1}$. Therefore,

$$\delta_{i,j} := r_{i,j} + \gamma v_{i,j+1} - v_{i,j},$$

and the GAE recursion is

$$A_{i,j}^{(\lambda)} := \delta_{i,j} + \gamma \lambda A_{i,j+1}^{(\lambda)}.$$

Normally $r_{i,j} = 0$ for interior tokens, so $\delta_{i,j} = \gamma v_{i,j+1} - v_{i,j}$.

For the final token of a nonterminal action span, the next decision is $a_{i+1,0}$ after observation o_{i+1} has been incorporated into the context. Thus,

$$\delta_{i,n_i-1} := R_i + \gamma v_{i+1,0} - v_{i,n_i-1}, \text{ and } A_{i,n_i-1}^{(\lambda)} := \delta_{i,n_i-1} + \gamma \lambda A_{i+1,0}^{(\lambda)}.$$

This is the well-defined version of the Hou et al. (2026) (4) and (5). The observation tokens are included in the next state, but no separate GAE steps are inserted for each observation token.

- Final token of a terminal episode. Suppose action span $M - 1$ terminates the episode. There is no next policy-decision state, so set $v_{\text{terminal}} = 0$ and $A_{\text{terminal}} = 0$. Then

$$\delta_{M-1,n_{M-1}-1} := R_{M-1} - v_{M-1,n_{M-1}-1},$$

and

$$A_{M-1,n_{M-1}-1}^{(\lambda)} := \delta_{M-1,n_{M-1}-1}.$$

This is the initialization point for the backward recursion in a completed episode.

- Truncated but nonterminal trajectory. A time limit, maximum-token limit, or context limit is not necessarily a true terminal state. Let s_{boot} be the valid policy-decision state from which the rollout would have continued, and define

$$v_{\text{boot}} := V_{\phi}(s_{\text{boot}}).$$

For the last sampled action token,

$$\delta_{\text{last}} := r_{\text{last}} + \gamma v_{\text{boot}} - v_{\text{last}}.$$

Since no later TD residuals were sampled, initialize the GAE carry as $A_{\text{next}} := 0$, giving

$$A_{\text{last}} := \delta_{\text{last}}.$$

The bootstrap value represents the unobserved future return. Setting $v_{\text{boot}} := 0$ for an ordinary truncation would incorrectly treat the truncation as failure or termination.

Actor advantage and critic target. SAO uses separate trace-decay choices for the actor and critic. Let

$$\lambda_{\text{policy}} := 1 - \alpha l^{-1}, \quad \lambda_{\text{critic}} := 1,$$

where l is the rollout length used by the estimator. Using the same TD residuals, compute two backward recursions:

$$A_{i,j}^{\pi} := \delta_{i,j} + \gamma \lambda_{\text{policy}} A_{\text{next}(i,j)}^{\pi},$$

and

$$A_{i,j}^V := \delta_{i,j} + \gamma \lambda_{\text{critic}} A_{\text{next}(i,j)}^V.$$

The actor uses $A_{i,j}^{\pi}$ in the policy objective. The critic target is

$$G_{i,j} := \text{stopgrad}(v_{i,j} + A_{i,j}^V),$$

and a typical value loss is

$$L_V = \frac{1}{2|\mathcal{A}|} \sum_{(i,j) \in \mathcal{A}} (V_{\phi}(s_{i,j}) - G_{i,j})^2,$$

where $\mathcal{A}$ is the set of model-generated action-token positions. Thus the value loss is defined only for states preceding model-generated action tokens. In the skip-observation formulation, the natural length is

$$l = \sum_i n_i,$$

the number of model-generated action tokens. Counting observation tokens would make tool-output verbosity change λ_{policy} , even though observation-token positions are excluded from the GAE recursion.

What “skip observation” means. Skipping observation tokens does not mean that the critic ignores observations. The critic evaluates

$$v_{i+1,0} = V_\phi(s_{i+1,0}),$$

and $s_{i+1,0}$ contains the observation returned after action a_i . Therefore, the observation directly affects both the next value prediction and the next policy distribution. What is skipped is only the insertion of individual observation-token positions into the GAE time axis:

- observation tokens are not policy actions;
- no policy advantages are assigned to observation tokens;
- no TD residuals are formed between consecutive observation-token positions;
- no GAE carry is repeatedly decayed through the observation’s token length;
- the last action token connects directly to the next generated action token.

Thus, the observation remains in the state, but observation-token positions are removed from the GAE recursion. The paper’s claim that this “filters out the stochasticity of environment feedback” should be interpreted carefully: the randomness and content of o_{i+1} still affect $s_{i+1,0}$, $v_{i+1,0}$, and $\pi_\theta(\cdot \mid s_{i+1,0})$. What the method removes is dependence on intermediate value predictions within the observation span, observation tokenization, observation length, and repeated $\gamma\lambda$ decay across externally generated tokens.

10.3 Practical Considerations

What to do?

- **Faster Value Update than Policy.** The authors attribute primary source of instability in single-rollout RL is the interdependence between the policy and the value function. If the value model V_ϕ is inaccurate, the advantage estimates $\hat{A}_\phi(s_t, a_t)$ become noisy, leading to destructive policy updates. Specifically, for every single gradient update applied to the policy π_θ , we enforce K updates to the value network V_ϕ (default $K = 2$).
- **Stabilizing Value Model Training via Parameter Freezing.** Instability of value model training, where the gradient norms of the value model are significantly larger than the corresponding policy model. Further decomposition shows that this instability originates primarily from the Full Attention layers. As a result, the authors freeze the parameters of the Full Attention layers during value model training.

What to monitor? A strong hybrid is often:

gross-staleness gate+PPO or soft rollback within the gate+explicit KL/TV monitoring.

For diagnosing whether the extra SAO masking is helping, the most informative measurements are not merely the aggregate clip fraction. Track the four A-by-r quadrants separately, the accepted-token fraction by actor version, importance-weight effective sample size, log-ratio quantiles, and the gradient norm contributed by the $A < 0, r > u$ quadrant. That last quadrant is where SAO is most likely to gain stability and where it sacrifices the strongest potentially corrective PPO signal.

Action Masking. Let the complete tokenized trajectory be $x_0, x_1, \dots, x_{T-1}$. Define the action mask as

$$M_p^{\text{action}} = \mathbf{1}\{x_p \text{ is generated by the policy}\}.$$

Suppose x_p is a generated action token, so $M_p^{\text{action}} = 1$. Its pre-action state is the prefix before that token,

$$s_p = x_{<p} = x_{\leq p-1}.$$

If the transformer value head emits

$$\bar{v}_t := V_\phi(x_{\leq t}),$$

then the value associated with action token x_p is

$$v_p := V_\phi(s_p) = V_\phi(x_{\leq p-1}) = \bar{v}_{p-1}.$$

Therefore, when the critic target for action token x_p is denoted by G_p , the physical-token value loss can be written as

$$L_V = \frac{1}{\sum_{p=1}^{T-1} M_p^{\text{action}}} \sum_{p=1}^{T-1} M_p^{\text{action}} (\bar{v}_{p-1} - \text{stopgrad}(G_p))^2.$$

Note that the index p starts at 1 because x_0 is not an action token (assuming non-empty context). Equivalently, reindex with $t = p - 1$:

$$L_V = \frac{1}{\sum_{t=0}^{T-2} M_{t+1}^{\text{action}}} \sum_{t=0}^{T-2} M_{t+1}^{\text{action}} (\bar{v}_t - \text{stopgrad}(G_{t+1}))^2.$$

Thus the implementation mask for critic outputs is a one-token-left shift of the action mask:

$$M_t^{\text{value}} = M_{t+1}^{\text{action}}, \quad t = 0, \dots, T - 2.$$

Cleaner implementation convention. The least confusing implementation is to shift logits, values, labels, targets, and masks into the same coordinate system. After the shift, every index refers to the same predicted action position, so the explicit $t + 1$ offset disappears from the loss code:

```

1  # Raw output at position t represents/predicts the state or token at t+1.
2  pred_logits = logits[:, :-1]
3  pred_values = raw_values[:, :-1]
4
5  target_tokens = input_ids[:, 1:]
6  policy_advantages = advantages[:, 1:]
7  value_targets = returns[:, 1:]
8
9  decision_mask = action_mask[:, 1:]
10 dis_mask = ratio_in_trust_region[:, 1:] # 1-eps_low < r < 1 + eps_high
11
12 policy_mask = decision_mask * dis_mask
13 value_mask = decision_mask
14
15 # The losses are conceptually
16 policy_loss = (
17     token_policy_loss(pred_logits, target_tokens, policy_advantages)
18     * policy_mask).sum() / policy_mask.sum().clamp_min(1)
19
20 value_loss = (
21     (pred_values - value_targets.detach()).square()
22     * value_mask).sum() / value_mask.sum().clamp_min(1)

```

Part IV

Appendix

A Glossary

A.1 Reward Model and Critic Model

Reward model. A model that scores a completed unit of behavior. In single-step LLM RL, the unit is usually the whole response y to a prompt x , and the reward model returns an outcome reward $r(x, y)$ only after the response is finished. It acts like the grader: the policy writes the answer first, then the reward model assigns the score being optimized.

In multi-step LLM RL, such as tool-use or agentic interaction, the completed unit may be an action span a_i , a full trajectory, or the final answer after several environment observations. The reward signal may therefore arrive after each completed action, as R_i , or only at the end of the episode. In either case, the reward model or verifier evaluates completed behavior, not a half-written prefix.

Critic model. A model that predicts the eventual return from the current partial state, usually written as a value function $V_\phi(s_t)$ over the prefix state s_t . Unlike the reward model, the critic does not grade the finished answer directly; it is a running estimate of what the future reward signal will imply for the current token position.

In single-step LLM RL, the critic looks at a partially generated response prefix and predicts the return induced by the final outcome reward $r(x, y)$, possibly together with token-level KL penalties or other shaping terms. In multi-step LLM RL, the critic evaluates states such as $s_{i,j}$ before generating token $a_{i,j}$, and predicts the return over future tokens, future action spans, environment observations, and later rewards. Thus the critic is a partial-credit forecaster for prefixes, while the reward model is the grader for completed behavior. The critic is trained by regression to a return target

A.2 How to train a Critic Model under the PPO framework

The critic V_ϕ is a separate network—typically the policy or reward model with the vocabulary head replaced by a scalar head—which emits one number per position in a single causal forward pass. It is defined by

$$V_\phi(s_t) = \mathbb{E}[r_t + \gamma r_{t+1} + \gamma^2 r_{t+2} + \dots | s_t], \quad (19)$$

i.e. the expected discounted reward from time t onward, *inclusive* of r_t .

A.2.1 TD residual

The temporal difference residual is defined as

$$\delta_t = \underbrace{r_t + \gamma V_\phi(s_{t+1})}_{\text{forecast after the step}} - \underbrace{V_\phi(s_t)}_{\text{forecast before the step}} \quad (20)$$

Read V_ϕ as a live win-probability meter. Before step t you believed $V_\phi(s_t)$; after taking the step you have banked r_t and can read the meter again. The difference is how much your outlook improved or worsened *because of this step*. Positive δ_t means better news than expected. δ_t measures how badly the critic contradicts itself between adjacent steps. A perfect critic has $\mathbb{E}[\delta_t] = 0$ everywhere. So δ does double duty—its expectation being nonzero trains the critic, its realization trains the actor.

A.2.2 The n -step ladder and GAE

Everything about GAE rests on the following cancellation. Substitute (20) into a discounted sum of residuals, writing V_k for $V_\phi(s_k)$, and take $t = 0$, $T = 3$ for concreteness:

$$\begin{aligned} \gamma^0 \delta_0 &= r_0 & + \gamma V_1 & - V_0 \\ \gamma^1 \delta_1 &= \gamma r_1 & + \gamma^2 V_2 & - \gamma V_1 \\ \gamma^2 \delta_2 &= \gamma^2 r_2 & + \gamma^3 V_3 & - \gamma^2 V_2 \\ \gamma^3 \delta_3 &= \gamma^3 r_3 & + \gamma^4 V_4 & - \gamma^3 V_3 \end{aligned} \quad (21)$$

Consider the value terms alone. The $+\gamma V_1$ of the first row cancels the $-\gamma V_1$ of the second; $+\gamma^2 V_2$ cancels $-\gamma^2 V_2$; $+\gamma^3 V_3$ cancels $-\gamma^3 V_3$. *Multiplying row ℓ by γ^ℓ is precisely what makes the powers line up*—this is the only reason the cancellation works. Two terms survive: $\gamma^4 V_4$, which is zero by the terminal condition $V_\phi(s_{T+1}) = 0$, and $-V_0$, which has nothing to cancel against. In general,

$$\boxed{\sum_{\ell=0}^{T-t} \gamma^\ell \delta_{t+\ell} = \underbrace{\sum_{\ell=0}^{T-t} \gamma^\ell r_{t+\ell}}_{\text{actual discounted return}} - V_\phi(s_t).} \quad (22)$$

In words: *the total surprise accumulated over a trajectory equals the outcome minus what you believed at the start*. The total is fixed; all GAE does is decide *when* along the trajectory that total gets booked.

Note carefully that (22) is not equation (26) below—it is that equation evaluated at $\lambda = 1$. The telescoping only goes through with weights γ^ℓ ; the general $(\gamma\lambda)^\ell$ case is built later.

Truncating (22) after n terms gives the **n -step advantage estimator**

$$A_t^{(n)} \equiv \sum_{\ell=0}^{n-1} \gamma^\ell \delta_{t+\ell}, \quad (23)$$

which interpolates between two extremes. At $n = 1$, $A_t^{(1)} = \delta_t$: minimum variance (only one step of randomness enters) but maximum bias, since you trust the critic’s forecast at $t+1$ completely. At $n = T - t + 1$ the sum runs to the end and (22) makes it the Monte Carlo advantage $(\sum_\ell \gamma^\ell r_{t+\ell}) - V_\phi(s_t)$: unbiased, since the critic appears only as a constant baseline, but maximum variance, because every random event after step t is now folded in—including events the token had no influence over.

GAE takes an exponentially weighted average over the whole ladder rather than picking one rung:

$$A_t^{\text{GAE}} = (1 - \lambda) \sum_{n \geq 1} \lambda^{n-1} A_t^{(n)}. \quad (24)$$

The weights $(1 - \lambda)\lambda^{n-1}$ sum to one, so this is a genuine average. Substituting (23) and exchanging the order of summation, the coefficient on $\delta_{t+\ell}$ is

$$(1 - \lambda) \gamma^\ell \sum_{n \geq \ell+1} \lambda^{n-1} = (1 - \lambda) \gamma^\ell \cdot \frac{\lambda^\ell}{1 - \lambda} = (\gamma\lambda)^\ell, \quad (25)$$

which recovers the familiar closed form

$$A_t^{\text{GAE}} = \sum_{\ell=0}^{T-t} (\gamma\lambda)^\ell \delta_{t+\ell}. \quad (26)$$

The parameter λ controls how far future TD residuals are allowed to flow back to the current token. At $\lambda = 0$, only the immediate residual δ_t is used, so the estimate is strongly bootstrapped from the critic: lower variance, but more bias if the critic is wrong. At $\lambda = 1$, every future residual is included with only the discount factor γ^ℓ , giving the Monte Carlo-style estimate: lower bias, but higher variance because all future randomness is credited backward. Intermediate values make this an exponential credit-assignment rule. In LLM RL, where one often sets $\gamma = 1$ to avoid washing out long completions, λ becomes the main knob deciding how much delayed reward from later tokens, later tool calls, or the final verifier should influence earlier token decisions.

A.2.3 The return target

The critic needs a regression target for (31) below. There is no unique choice; there is a ladder, indexed by how many real steps you take before letting the critic guess the remainder:

$$\hat{R}_t^{(n)} = \underbrace{r_t + \gamma r_{t+1} + \cdots + \gamma^{n-1} r_{t+n-1}}_{n \text{ steps of observed reward}} + \underbrace{\gamma A^{(n)} V_\phi(s_{t+n})}_{\text{critic guesses the tail}}. \quad (27)$$

At $n = 1$ this is the plain TD target $r_t + \gamma V_\phi(s_{t+1})$; at $n = T - t + 1$ the bootstrap term vanishes and it is the full observed return.

The key observation is that *each rung of this ladder differs from the matching rung of the advantage ladder by exactly $V_\phi(s_t)$* . Subtracting $V_\phi(s_t)$ from (27) and telescoping exactly as in (21) yields $\hat{R}_t^{(n)} - V_\phi(s_t) = A_t^{(n)}$, i.e.

$$A_t^{(n)} = \hat{R}_t^{(n)} - V_\phi(s_t) \iff \hat{R}_t^{(n)} = A_t^{(n)} + V_\phi(s_t). \quad (28)$$

Return targets and advantages are **the same object shifted by the baseline**—and crucially, the shift $V(s_t)$ is *the same for every rung*: it carries no n .

Apply the averaging operator $\sum_{n \geq 1} (1 - \lambda) \lambda^{n-1} (\cdot)$ to both sides. On the left it produces the λ -return by definition; on the right it splits in two, and because $V_\phi(s_t)$ is constant in n it factors straight out of the second sum:

$$\underbrace{\sum_{n \geq 1} (1 - \lambda) \lambda^{n-1} \hat{R}_t^{(n)}}_{\hat{R}_t^\lambda} = \underbrace{\sum_{n \geq 1} (1 - \lambda) \lambda^{n-1} A_t^{(n)}}_{A_t^{\text{GAE}} \text{ by (24)}} + V_\phi(s_t) \underbrace{\sum_{n \geq 1} (1 - \lambda) \lambda^{n-1}}_{= 1}. \quad (29)$$

The leftover weight sum is exactly 1—this is where it matters that (24) is a genuine *average* and not an arbitrary linear combination—so the baseline passes through untouched and

$$\hat{R}_t \equiv \hat{R}_t^\lambda = A_t^{\text{GAE}} + V_\phi(s_t). \quad (30)$$

Note that no particular n has been selected here; the identity holds because *every* rung shares the same baseline. The bare $\hat{R}_t$ drops the λ superscript because this blended quantity is *the* target: the one equation (31) regresses against. So (30) is not a new estimator—it is a rearrangement. There is one blended estimate of how good s_t turned out to be. Viewed as a *deviation from the critic’s prediction* it is the advantage that drives the actor; viewed as an *absolute number* it is the target the critic regresses toward:

$$\mathcal{L}^{\text{VF}}(\phi) = \mathbb{E}_t \left[(V_\phi(s_t) - \hat{R}_t)^2 \right]. \quad (31)$$

Informally: $V_\phi(s_t)$ is the prior, A_t^{GAE} is the accumulated (γ -discounted, λ -decayed) news that arrived afterward, and the sum is the posterior estimate you now wish the critic had produced.

So is the critic minimizing the advantage?

Substituting (30) into (31) gives $V_\phi(s_t) - \hat{R}_t = -A_t^{\text{GAE}}$, hence

$$\mathcal{L}^{\text{VF}}(\phi) = \mathbb{E}_t \left[(A_t^{\text{GAE}})^2 \right]. \quad (32)$$

The critic loss *is* the squared advantage. Two qualifications keep this from misleading.

The target is detached. $\hat{R}_t$ is computed once from the *old* critic, before the optimization epochs, and frozen; gradients flow only through the fresh $V_\phi(s_t)$ on the left. So (32) holds numerically at the first inner epoch, where $\phi = \phi_{\text{old}}$, and thereafter the loss is a regression against a stale constant.

```

1  with torch.no_grad():                                # <-- everything here is frozen
2      values      = critic(states)                      # V_{phi_old}
3      advantages  = compute_gae(rewards, values, gamma, lambda)
4      returns     = advantages + values                 # hat{R}, a constant tensor
5
6  for epoch in range(K):                                # PPO inner epochs
7      v = critic(states)                                # only THIS carries gradient
8      vf_loss = ((v - returns) ** 2).mean()
```

What matters is the gradient,

$$\nabla_\phi \mathcal{L}^{\text{VF}} = 2 \mathbb{E}_t \left[(V_\phi(s_t) - \hat{R}_t) \nabla_\phi V_\phi(s_t) \right] = -2 \mathbb{E}_t \left[A_t^{\text{GAE}} \nabla_\phi V_\phi(s_t) \right], \quad (33)$$

which says: *the advantage is the critic’s error signal*. Positive advantage means the state turned out better than predicted, so raise $V_\phi(s_t)$ by an amount proportional to $|A_t|$. This is exactly the forward view of TD(λ), $\Delta\phi \propto (\hat{R}_t^\lambda - V_\phi(s_t)) \nabla_\phi V_\phi(s_t)$, rewritten as a squared error so that autodiff performs the update. Letting gradients flow through the bootstrapped $V_\phi(s_{t+n})$ inside $\hat{R}_t$ instead yields Baird’s residual-gradient method—a different algorithm with a different fixed point, which invites the critic to reduce loss by dragging $V_\phi(s_{t+n})$ toward $V_\phi(s_t)$, satisfying self-consistency by chasing its own tail rather than by becoming accurate. Semi-gradient (detached target) is standard. At $\lambda = 1$ the distinction is moot, since $\hat{R}_t$ is the observed return and contains no ϕ .

Minimizing $\mathbb{E}[A^2]$ does not drive A to zero. The minimizer of a squared error is the conditional mean, $V^*(s) = \mathbb{E}[\hat{R}_t \mid s_t = s]$, and the residual there is the conditional variance $\text{Var}(\hat{R}_t \mid s_t)$ —irreducible however good the critic becomes. A perfect critic yields $\mathbb{E}[A_t \mid s_t] = 0$, not $A_t = 0$.

There is a structural reason it cannot collapse: $V_\phi(s_t)$ never sees the action a_t . It is evaluated on the prefix *before* the token is sampled, so it can represent only the average over actions under the current policy. The action-specific deviation $A_t = Q(s_t, a_t) - V_\phi(s_t)$ is precisely the component the critic is architecturally forbidden to absorb. It strips out what is predictable from the state and leaves untouched the question of which token was actually chosen.

A.2.4 Implementation notes

Backward recursion. Equation (26) is never evaluated as a sum. Expanding it one term gives the recursion actually used,

$$A_t = \delta_t + \gamma\lambda A_{t+1}, \quad A_{T+1} = 0, \quad (34)$$

computed in a single backward pass—“my advantage is my own surprise, plus a discounted share of everything that surprises the future.” Equation (30) is then one addition, `returns = advantages + values`, which is the real reason the target is written in that form.

Normalization ordering. Advantages are usually normalized to zero mean and unit variance before the policy loss. The return targets must be computed *before* that normalization, or the critic is trained toward a rescaled quantity. This is a common and quiet bug.

Discounting at token granularity. A 20k-token rollout with $\gamma = 0.99$ discounts the terminal reward by roughly e^{-200} , annihilating the signal. Hence $\gamma = 1$ is effectively universal in LLM RL, which leaves λ carrying the entire bias–variance tradeoff by itself.

Masking observations. Failing to mask environment tokens out of the policy loss trains the model to *predict tool outputs*, so it learns to hallucinate plausible observations and answer from them. The environment gets distilled into the policy and rewarded for confabulating.

Stochastic transitions at tool boundaries. In single-turn LLM RL the transition is deterministic— s_{t+1} is s_t with your own token appended—so all randomness lives in the policy. At a tool boundary the transition is genuinely stochastic: the same query may return an excellent document or garbage. The residual across that boundary, $\delta_t = \gamma V_\phi(s_t \oplus \text{obs}) - V_\phi(s_t)$, mixes skill (a well-formed query) with luck (a cooperative index), and PPO cannot separate them. At $\lambda = 1$ that luck propagates undiminished back onto every earlier token, which is a concrete argument for $\lambda < 1$ —or for turn-level value baselines—in agentic training.

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